

Section A (40 marks)

Answer all questions in the space provided

Question 1

Fig. 1.1 shows a cell undergoing two different stages of meiosis (to the same scale).

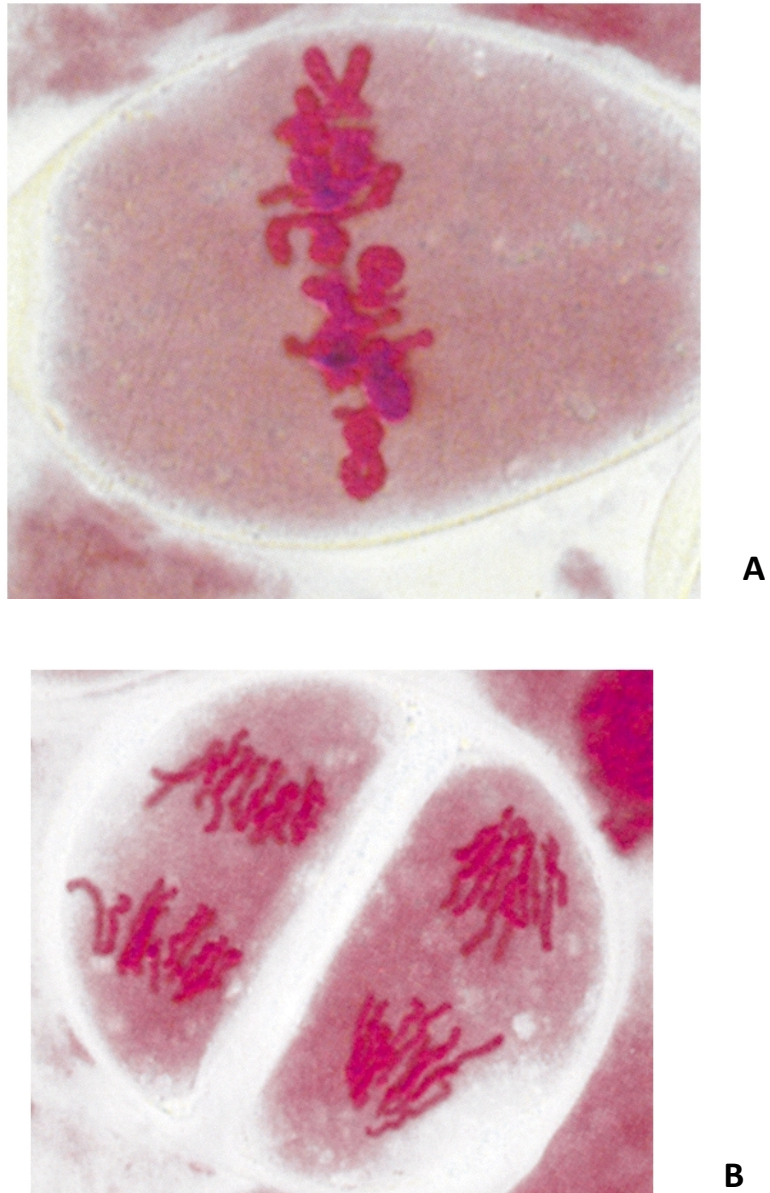


Fig. 1.1

(a) Identify stages **A** and **B**. [1]

- A is **metaphase I**
- B is **anaphase II**.

(b) Explain the phenomenon observed in B. [2]

- **Centromeres divide**
- and the **chromatids separate** and are pulled, centromeres first, by the spindle fibres **to opposite poles**.
- This is due to the **shortening of** the centromere-to-pole **spindle fibres**.

[1 mark for any of the 2 points above]

(c) Describe how meiosis can lead to genetic variation. [2]

- In **Prophase I**, non-sister chromatids of **homologous chromosomes** align themselves alongside one another, and **form chiasma**. **Crossing over** occurs and new combinations of alleles on the chromosomes of the gametes are formed.
- During **Metaphase I** and **Anaphase I** respectively, the **arrangement and subsequent separation of the chromosomes** of each bivalent is completely **independent of** the orientation of the **other bivalents**. This gives rise to different combinations of chromosomes in a gamete.
- During **Metaphase II** and **Anaphase II** respectively, the **arrangement and subsequent separation of the chromatids** of each chromosome is **completely independent of** the orientation of the **other chromosomes**. This gives rise to different combinations of chromosomes in a gamete.

[1 mark for any of the 2 points above]

Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

(d) Suggest a possible outcome of a disorganised metaphase plate. [1]

- **The chromosomes and chromatids may not separate equally/properly during Anaphase I or II respectively. [0.5]**
- **Meiosis cannot continue and no germ cells are formed / This lead to non-disjunction. [0.5]**

OR

- **The sister chromatids may not separate equally/properly during Anaphase. [0.5]**
- **Mitosis / cell division cannot continue and no daughter cells are form. [0.5]**

The chromosome carrying alleles **F/f** also carries alleles **E/e**, as shown in **Fig 1.2**.

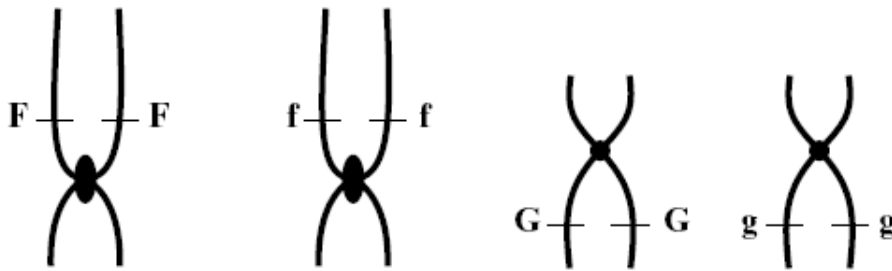


Fig. 1.2

(e)(i) State the genotype of this cell. [1]

FfGg

(ii) Explain how there are two copies of each allele of each gene at this stage of nuclear division. [2]

- DNA replication has occurred;
- During interphase / S-phase of interphase before mitosis;
- Giving rise to two chromatids;
- With two copies of the allele, one on each chromatid;

[Total: 9 marks]

Question 2

Fig. 2.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

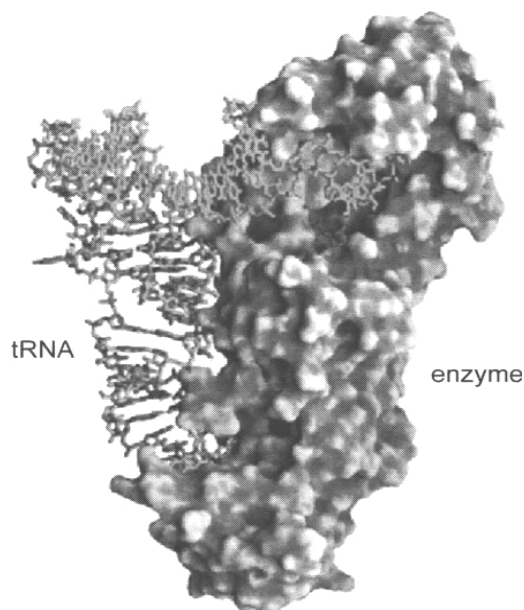


Fig. 2.1

(a) Describe **two** structural features which adapt tRNA to its role in translation. [2]

1 3' / (CCA end) = allows specific amino acid molecule binds to the tRNA

2 loop with anticodon /region has specific triplet base sequence - complementary to a codon on mRNA.

(b) Explain how the enzyme in **Fig. 2.1** is suited to its function. [3]

1. Identify the enzyme as aminoacyl-tRNA synthetase
2. has recognition/active site for both amino acid
3. and tRNA with specific anticodon;
4. with site being complementary in shape and charge to the substrates
5. able to catalyse formation of covalent bond between tRNA and amino acid
6. twenty different aminoacyl-tRNAase enzymes for each amino acid to ensure specificity in binding

(c) **Fig. 2.2** below shows the base sequence on a length of mRNA. The key indicates which amino acids they represent.

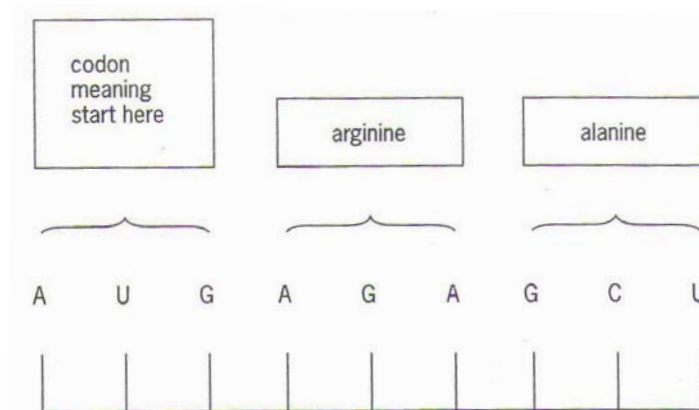


Fig. 2.2

The amino acid arginine is shown here as AGA but can also be represented by the codon AGG.

(i) Explain why the substitution of A by G in this case does not affect the amino acid represented. [1]

Genetic code is degenerate/redundancy in genetic code;

1 amino acid can be coded by more than 1 codon;

First two bases are most important in determining which amino acid is attached;

The third base is a wobble base/not as important;

It is a form of silent mutation

1 max

- (ii) The codons in the mRNA sequence do not overlap. Suggest 2 advantages of this. [2]

Substitution of a base will only change one amino acid instead of two;;
Overlapping can cause difficulties in translation when the tRNA comes into the P site as it would be blocked by the previous tRNA;;
There is less restriction on the order of amino acids since the overlapped base of one codon has no effect on the next codon;;

Max 2

Some genetic drugs are short sequences of nucleotides. They act by binding to selected sites on DNA or mRNA molecules and preventing the synthesis of disease-related proteins. There are two types:

1. triplex drugs are made from DNA nucleotides and bind to the DNA forming a three stranded helix.
2. antisense drugs are made from RNA nucleotides and bind to mRNA.

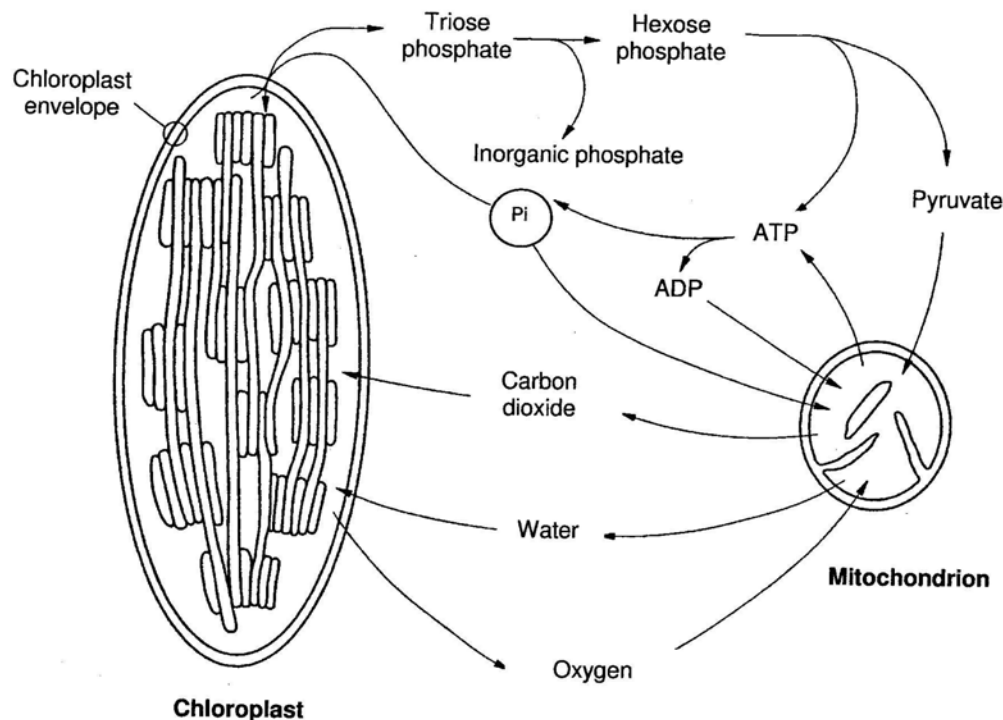
- (iii) Name the process in protein synthesis which will be inhibited by [1]:

Drug type	Process inhibited
Triplex drugs	Transcription
Antisense drugs	Translation

[Total: 9 marks]

Question 3

Fig. 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.



A leafy shoot is sealed inside an air-tight container. The concentration of oxygen in the atmosphere within this container is measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 3.1**, outline how the following links the metabolic pathways between the chloroplast and mitochondrion of this cell.

- (i) triose phosphate and hexose phosphate; [2]

in stroma of chloroplast, triose phosphate (TP) is formed as a product of light independent / dark reaction;(1m) TP moves out of chloroplast into the cytosol and is converted to hexose phosphate (1m)

that can enter glycolysis as a respiratory substrate, to be eventually oxidized during aerobic respiration into pyruvate (1m) which enters the mitochondria to undergo the link reaction / to be converted to acetyl CoA(1m) max [2]

- (ii) gaseous exchange. [2]

In granum of chloroplast, oxygen is formed as a product of photolysis of water during the light dependent/ light reactions (1m); oxygen diffuse out of chloroplast and act as the final electron acceptor in the electron transport chain (ETC) in mitochondrial matrix(1m); CO₂ released during Krebs cycle/ link reaction released into cytosol which diffuse into stroma of chloroplast and used in dark / light independent reaction/ carbon fixation (1m)
max [2]

- (b) With reference to **Fig. 3.1**,

- (i) explain how it is possible for the oxygen concentration to remain constant. [1]

Oxygen released by chloroplast during the light dependent reaction of photophosphorylation/ photolysis of water is used by mitochondrion as final electron acceptor, compensation point has been reached(1m) Rate of respiration equal to rate of photosynthesis(1m), thus no net change in oxygen concentration.(1m)
max [1]

- (ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism? [2]

ATP derived from mitochondria / oxidative phosphorylation are supplied to the cell to meet its need(1m) ; triose phosphate from light-independent/ dark reaction will move to cytosol and may eventually be used as respiratory substrate(1m) to drive glycolysis and thus generate ATP.(1m)
max [2]

In plants, the carbohydrates manufactured are channeled to form two types of macromolecules, namely cellulose and starch.

- (c) State the two structural differences between cellulose and starch. [2]

- Cellulose basic units β glucose while starch is composed of α glucose
- Cellulose is composed of straight chains polysaccharide while starch made of polysaccharide chains that are coiled into helices
- Cellulose has hydrogen bonding between adjacent chains while no such hydrogen bonds exist between the polysaccharide chains.
- Cellulose polysaccharide chains are unbranched while in starch (amylopectin), there is branching (glycosidic bond is between carbons 1 and 6)

1 mark per point/ max [2]

- (d) Explain how these differences allow them to serve different functions in a plant cell. [2]

Cellulose: long straight chains allow for stacking and formation of hydrogen bonds to form bundles of fibrils to give tensile strength (1m) which is necessary in plant cells to give support and structure (1m).

Starch is coiled and compact to serve as storage material (1m), occupying less space and is insoluble so it does not affect osmotic potential of cell, (1m)

1 mark per point/ max [2]

[Total: 11 marks]

Question 4

In 1905, the Dutch botanist Hugo De Vries, in an experiment to test Mendel's rediscovered work, found that when a red-flowered, three-leaved clover plant was crossed with a white-flowered, five-leaved plant the following offspring were produced:

36 red-flowered, five leaved
41 red-flowered, three-leaved
40 white-flowered, five-leaved
38 white-flowered, three-leaved

The alleles for red flowers and five leaves are dominant.

(a) Using the symbols **R** for red flowers, **r** for white flowers, **E** for five leaves and **e** for three leaves, draw a genetic diagram to explain these results. [5]

(b) When these plants were self-pollinated, explain why only the white flowered, three-leaved plants bred true. [2]

[Total: 7 marks]

- 4 (a) Using the symbols **R** for red flowers, **r** for white flowers, **E** for five leaves and **e** for three leaves, draw a genetic diagram to explain these results. [4]

parental phenotype	red-flowered		white-flowered	
	3-leaved		5-leaved	
parental genotype	Rree		rrEe;	
gametes	Ⓔ	ⓇⓇ	Ⓔ	ⓇⓇ;
	Punnett sq.;			
Offspring genotype	RrEe :	Rree :	rrEe :	rr ee
Offspring phenotype	red	red	white	white
	5-leaved	3-leaved	5-leaved	3-leaved
Ratio	1	:	1	:
	1	:	1	:
	1	:	1	:
	1	:	1	:

- (b) When these plants were self-pollinated, explain why only the white-flowered, three-leaved plants bred true. [2]

homozygous recessive;
produce only 1 type of gametes;

Question 5

A new brightly-coloured mouse was discovered in Russia and it was given the name *Mus colores*. **Fig. 5.1** below shows two flowcharts that illustrate how blue and green cells were isolated from different parts of this mouse, and each cell was cultured individually.

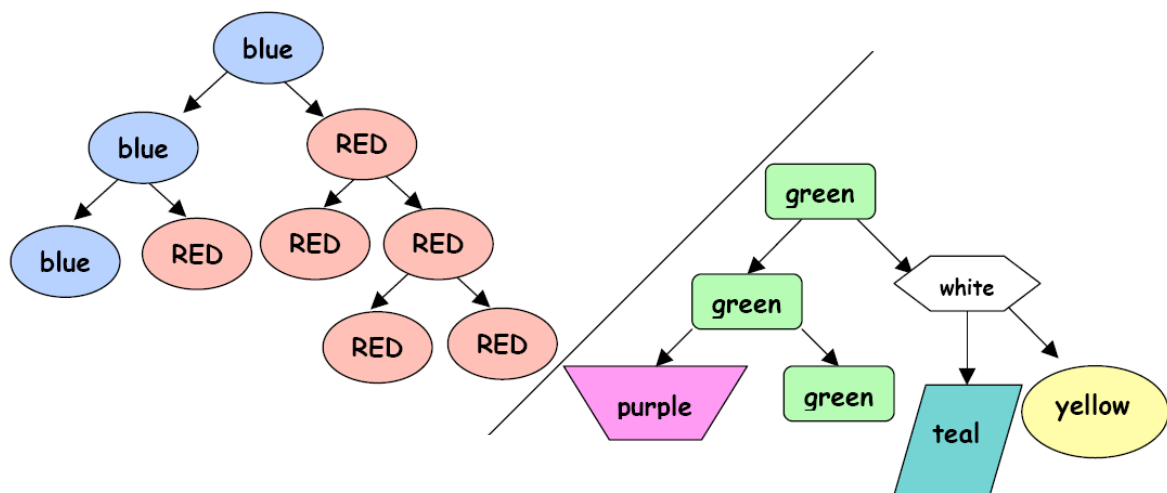


Fig. 5.1

- (a) With reference to **Fig. 5.1**,
- (i) are the blue cells stem cells? Give a reason for your answer. [1]

Yes as they are **capable of self-renew/ giving rise to other cell types**.

- (ii) given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]

-Green cell is **multipotent**: can **give rise to more than one cell type**.
-They can be **influenced** by the **presence of internal** and **external signals**

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 5

- (a) Explain, with examples, how genetic engineering can improve the quality and yield of crop plants and animal. [14]

quality of crops

-Quality of crops can be increased by introducing herbicide resistant, aluminium tolerant, cold tolerant, Vitamin A etc genes into the plants.

Different ways of introducing the genes include

-using a gene gun which involves shooting DNA-coated tungsten or gold particles into plant cells.

- electroporation which uses a short burst of electricity to get the DNA into the cell.

- *Agrobacterium tumefaciens* - mediated transformation

- *A. tumefaciens* is a soil bacterium and contains a plasmid called the Ti plasmid.

-Two components of the Ti plasmid, the T- DNA and the *vir* region (for virulence) , are essential for the transformation of plant cells.

-vir region will mediate the transfer of T-DNA.

- Hence, the T-DNA (Transferred DNA) can be excised from the Ti plasmid, transferred and integrated to a plant cell.

Details of Agrobacterium transfer

-The plasmid and a fragment containing the foreign gene are treated with the same restriction enzymes separately.

- Sticky ends are formed and the foreign gene can be inserted into the plasmid when mixed together, forming recombinant plasmid.
- The recombinant plasmid is then reinserted back into the bacteria, producing recombinant bacteria which are used to infect the plant.
- The common choice of tissue are immature embryos, leaf disks and apical meristems.
- As the plant genome is expressed, the foreign gene will also be expressed.
- The transformed plant cells can be identified using media containing antibiotics or herbicides.

Improve yield

- Use of *E. coli* and microinjection of DNA into fertilised egg

Use of E. Coli

Example of somatotrophin

- Somatotrophin is injected into cows to enhance milk production.
- Somatotrophin is mass produced by genetic engineering.(cloning)

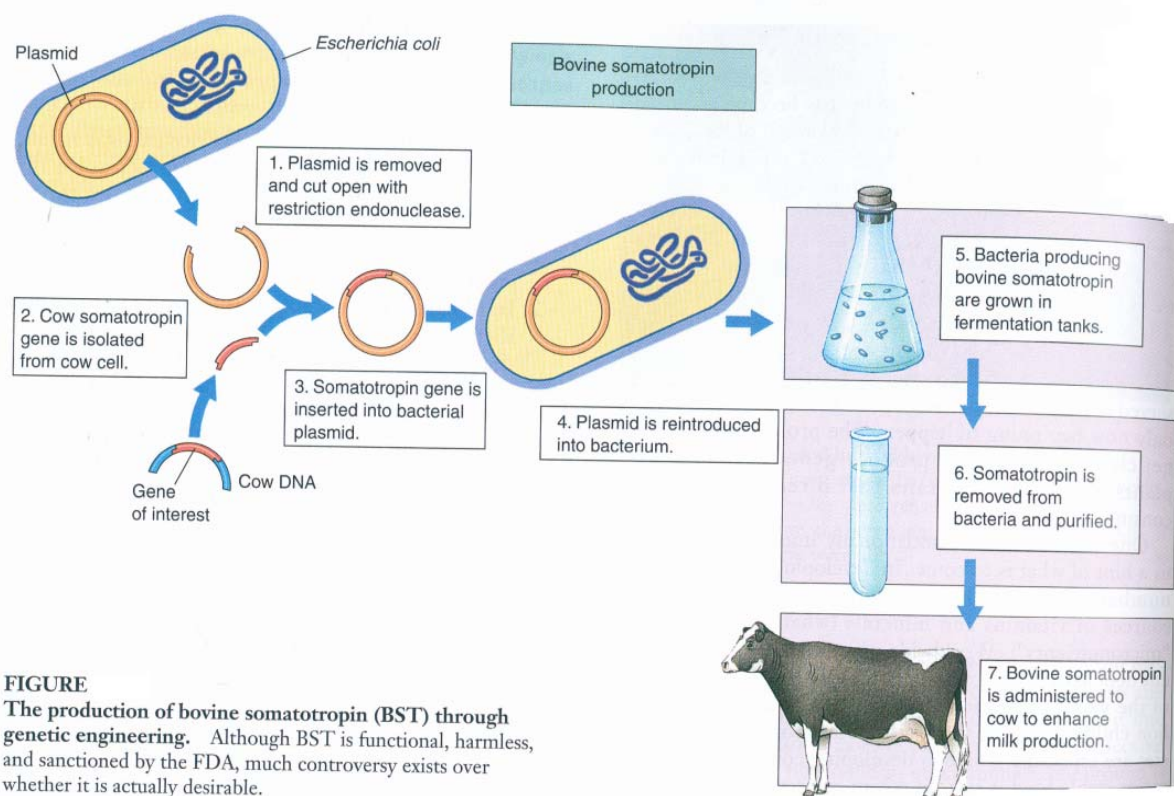


FIGURE
The production of bovine somatotropin (BST) through genetic engineering. Although BST is functional, harmless, and sanctioned by the FDA, much controversy exists over whether it is actually desirable.

Microinjection of DNA into fertilised egg

-Injecting enhanced growth hormone levels might result in leaner pigs with improved meat quality and with faster growth. Other scientists introduced growth hormone transgenes into fish and chickens with similar objectives.

1. Produce recombinant DNA by combining transgene in a bacteria plasmid.

2. Prior to microinjection, the eggs are surgically removed from the female parent.

3. Egg is fertilized with sperm *in vitro*.

4. Before the male pronucleus can fuse with the female pronucleus, DNA microinjected into the male pronucleus through a very fine tipped glass needle 5. Usually, several hundred to several thousand copies of the gene of interest are injected into each egg, and multiple integrations often occur in the chromosomal DNA.

-Because the DNA is injected into the fertilized egg, integration of the injected DNA molecules usually occurs early during embryonic development.

- as a result, some germ-line cells may carry the transgene. As would be expected, the animals that develop from the injected eggs called the G_0 generation-are almost always genetic mosaics, with some somatic cells carrying the transgene and others not carrying it.

- initial (G_0) transgenic animals must be mated together and G_1 progeny produced to obtain animals in which all cells carry the transgene.

(b) Describe the properties of plasmids that would allow them to be use as vectors. [6]

1. Small;
 - Enable it to enter host cells easily;
2. Contains an origin of replication;
 - → gene of interest can be replicated autonomously in a host cell; OR
 - ensures plasmid is not lost during subsequent host division;
3. Can use high copy number plasmid;
 - Enable multiple copies of gene of interest to be synthesized per cell;
4. Contains selectable markers to select for transformants with recombinant plasmids;
 - Eg (those coding for antibiotic resistance / nutritional markers;
 - 1st round of selection: to select for presence of plasmid; eg;
 - 2nd round of selection: to select for presence of recombinant plasmid; eg;
5. Restriction sites located at a selectable marker;
 - Plasmid is cut and gene inserted → insertional inactivation;

Question 6

(a) Describe and explain the secondary structure of proteins. [6]

- 1 derived from the primary sequence of amino acids
- 2 Secondary structures are either of **α helices**; **β -pleated sheets**;
- 3 Shape is maintained by **weak hydrogen bonds**.
- 4 H-bonds are **intra molecular /intra chain** for **α -helices** between every **forth amino acid/aa 1 with aa 5**; comprises
- 5 a weakly **positive H atom in the –NH gp** and a **weakly negative oxygen atom in the -CO** ;
- 6 **β -pleated sheets** can be arranged in a **parallel or anti-parallel orientation**;
- 7 comprise of weak **intra chain** H bonding between **positive H atom in the –NH gp** and a **weakly negative oxygen atom in the -CO**

(b) Describe the main ways in which a globular protein differs from DNA. [6]

Pt of comparison		Globular proteins	DNA
1	Shape	Compact & round globular	Linear
2	Solubility	Yes	No/less soluble
3	Monomers	Amino acids	Nucleotides
4	Types of monomers	20 used in humans	4 types (A, T, G, C)
5	Bonding 1	Linked by peptide bonds	Linked by phosphodiester bonds
6	Number of subunits	One or more	Always 2 under physiological conditions
7	Bonding 2	Polypeptide chains/subunits linked by ionic bonds, covalent disulphide bonds, hydrophobic interactions and hydrogen bonding	Only hydrogen bonding links the two DNA chains
8	Function	Multiple functions: Enzymatic, structural, transport, energy source (any 2)	Carry hereditary information/code for synthesizing products
9	What confers functionality	Its function is usually linked to its 3D shape/ conformation and configuration	Its function is dependent only on the order in which the monomers are arranged in DNA chain
10	Levels of organization	4 levels	1 structure/ α –helix
11	Charge	Can be of any charge	Negatively charged
12	Location	Everywhere/both inside and outside the cell	Only inside the cell; R DNA only in nucleus (found in cytoplasm)

(c) Compare and contrast between DNA replication and transcription. [8]

Similarities:

1. Both involve unwinding of the double helix DNA;;
2. Both involve breaking of the weak hydrogen bonds between complementary bases;;
3. Both involve formation of phosphodiester bonds between adjacent nucleotides;;
4. Both involve aligning free nucleotides through complementary base pairing;;

5. Both products elongate in 5' to 3' direction::

6. Both occur in the nucleus::

[max 5 marks]

Differences:

Features	DNA replication	Transcription
1. (Main) Enzyme involved	<u>DNA polymerase III</u> - binds to the parental DNA molecule at the origin of replication;	<u>RNA polymerase II</u> - binds to the promoter, a region near the beginning of the gene to be transcribed;
2. Raw Material	<u>Deoxyribonucleotides</u> ;	<u>Ribonucleotides</u> ;
3. Template	<u>Both strands</u> of DNA molecule act as a template;	<u>One strand</u> of DNA molecule acts as the template;
	Involves <u>entire sets of chromosomes</u> ;	Involves <u>only a few genes</u> ;
4. Base Pairing	<u>Adenine with Thymine</u> and <u>Cytosine with Guanine</u> ;	<u>Adenine with Uracil</u> and <u>Cytosine with Guanine</u> ;
5. Presence of proofreading property on enzyme involved	<u>DNA polymerase proofreads</u> the newly synthesized daughter strand, ensuring precise complementary base pairing;	<u>RNA polymerase II</u> has no proof-reading function;
6. Products	<u>2 DNA molecules</u> ;	<u>1 RNA molecule</u> ;
7. Location of the products	Products <u>remain in the nucleus</u> ;	Products <u>leave the nucleus into cytoplasm</u> ;
8. Post – transcriptional modification	Products <u>do not go through post - transcriptional modification</u> ;	Products / pre-mRNA may go through <u>post-transcription modification</u> / splicing / capping / poly-A tailing;
9. RNA priming	RNA primers are synthesized by primase to initiate DNA synthesis;	Does not require RNA priming;
10. Direction of elongation	The elongation of DNA occurs in a <u>bidirectional manner on both templates</u> ;	The elongation of mRNA occurs <u>in 5'→3' direction</u> on <u>single template</u> ;
11. Synthesis	Discontinuous on lagging stand and continuous on leading strand;	Continuous;
12. When the process occur	Prior to nuclear division / mitosis / meiosis;	Prior to translation;